Soyesh Shrestha
STUDENT-ID : 230054
Coventry ID: 14188905
BSc. Hons Computing, Softwarica College of IT and E-Commerce,
Coventry University
ST5014CEM Data Science for Developers
Siddhartha Neupane
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Abstract

This project utilizes techniques of preprocessing, refinement, and extraction of meaningful information out of raw datasets through data science techniques. In its first stage, preprocessing of data involved checking for discrepancies, gaps, and outliers, and resolving them for increased accuracy and dependability of information. Techniques such as imputation, transformation, and normalization were utilized through R programming for preprocessing of the dataset for analysis.Data visualization then helped in extraction and visualization of meaningful information effectively. With packages such as ggplot2, dplyr, and tidyverse in R, cleaned information was reorganized in concise, pleasing-to-the-eye plots and charts. Through visualization, trends, patterns, and relationships that could have been challenging for stakeholders to perceive in raw and complex forms of information could be discovered and understood.

The project brings out the necessity of cleaning in data science and proves how visualization aids in decision-making through information. It is a powerful demonstration of the use of R in resolving real-life information-related issues through organized analysis and out-of-the-box thinking.

Keywords


Data Science
Devon
Data
DATA
Postcode
Data Vicieation
Visuallation
Data Science
DATA SCENCR
R
VISUALZATION
VISUALATION
POSTCOST
Dorvost
CRIMY
Crime
CRIME
County
Postcode

Table of Contents

Introduction

This project discusses the application of data science techniques in analyzing and extracting insights from datasets that contain house prices, broadband speeds, crime rates, and school performance metrics.
The main objective of this project was to convert raw, unstructured data into useful information by cleaning the data, linear modeling, and making recommendation systems. The project began with a rigorous data preprocessing approach to eliminate inconsistencies, missing values, and noise in the provided datasets Cleaning of the data turned out to be very essential for the accuracy and reliability of any further analyses.

Linear modeling was then used to identify relationships and dependencies between variables, such as a correlation of house prices and broadband download speeds, or impact on demands due to crime rates on housing.It resulted in the development of the recommendation system. With the key insights derived from the analyses, the system plotted a list of top 10 towns by their factors, where suggestions could be made for decisions.It is an infrastructure project which shows how data cleaning, statistical modeling, and visualization are merged in view of solving a real-world issue.The above process, therefore, highlights the value of data science in discovering relevant insights and resolving complicated multidimensional datasets.

Data cleaning

Data cleaning is the process that ensures the datasets are right and useful in the analysis to be done. The raw data in this project was obtained, involving house prices, broadband speeds, crime rates, and school performance, which were characterized by missing values, inconsistency, and outliers. For all these problems, I used R programming.

R programming: to impute or remove missing values, detect, and treat outliers, and normalize data wherever required. Various R libraries such as dplyr for the preparation and cleaning of the data were employed in this process. Data cleaning is crucial in making sense from raw data and ensuring the reliability of the results.

House Price cleaning code

```r
# Load required libraries for data handling and cleaning
library(tidyverse)  # Includes packages for data analysis and visualization
library(dplyr)      # Provides functions for data manipulation
library(readr)      # Enables reading and writing of CSV files

# Define column names for datasets without headers
column_names <- c(
  "ID", "Price", "Date", "Postcode", "Type", "NewBuild", "Tenure_Type",
  "HouseNumber", "ApartmentNumber", "Street", "Locality", "Town",
  "District", "County", "Category", "Record_Status"
)

# Import datasets for the years 2021, 2022, and 2023, specifying no column headers
data_2021 <- read_csv("C:/Users/soyes/Downloads/Coventry ID -14188905/Obtain data/House price/pp-2021.csv"
data_2022 <- read_csv("C:/Users/soyes/Downloads/Coventry ID -14188905/Obtain data/House price/pp-2021.csv"
data_2023 <- read_csv("C:/Users/soyes/Downloads/Coventry ID -14188905/Obtain data/House price/pp-2021.csv"

# Assign the defined column names to each dataset
colnames(data_2021) <- column_names
colnames(data_2022) <- column_names
colnames(data_2023) <- column_names

# Combine the data for all three years into a single dataset
combined_data <- rbind(data_2021, data_2022, data_2023)

# Clean and filter the combined dataset
cleaned_data <- combined_data %>%
  select(Postcode, Price, Date, County, Town) %>%  # Retain only relevant columns
  filter(County %in% c("DEVON", "DORSET")) %>%    # Filter rows to include only Devon and Dorset
  na.omit()                                       # Remove rows with missing data

# Save the cleaned dataset to a CSV file
write_csv(cleaned_data, "clean data/houseprice.csv")
```

This code is going to facilitate the cleaning and merging of house price data placed in three CSV files for the years 2021, 2022, and 2023. Loading neat libraries/lubridate/dplyr to manipulate the data and handle dates. Each of these CSV files will be read; each needs to be combined with the corresponding one of the other years to turn into a single dataset. The column names in data for the years 2022 and 2023 are henceforth harmonized with the corresponding column names in 2021 for clarity; then the name of the unified dataset is also changed. In order to have an initial view of the combined dataset, check its structure first and convert the column 'Transaction_Date' into a proper date format for further ease of analysis.

The next line of code after this filters the data for the houses in the counties of Devon and Dorset, keeping only the relevant columns of postcode, price, and transaction date, after which subsets are exported to individual CSV files. The filtered data for the counties of Devon and Dorset are then jointly combined into one dataset, which is then exported into a final CSV file. This further has a clean data set where merged house-price data for the named counties can be subjected to further analyses.

Broadband cleaning code

```r
# Load required libraries for data handling and manipulation
library(tidyverse)  # Includes ggplot2, dplyr, tidyr, readr, and more
library(dplyr)      # For efficient data manipulation
library(readr)      # For reading CSV files

# Import the broadband dataset
broadband_data <- read_csv("C:/Users/soyes/Downloads/Coventry ID -14188905/Obtain data/Broadband/201809_fixe

# Import the housing price data for 2021
housing_price_2021 <- read_csv("C:/Users/soyes/Downloads/Coventry ID -14188905/clean data/houseprice.csv")

# Convert columns to the same data type for accurate merging
broadband_data$postcode_space <- as.character(broadband_data$postcode_space)
housing_price_2021$Postcode <- as.character(housing_price_2021$Postcode)

# Merge broadband data with town and county details from the housing price dataset
broadband_data_with_town <- broadband_data %>%
  left_join(
    housing_price_2021 %>% select(Postcode, Town, County),  # Select only relevant columns
    by = c("postcode_space" = "Postcode")                 # Match based on postcode
  )

# Refine and filter the merged broadband data
broadband_clean_data <- broadband_data_with_town %>%
  select(
    Postcode = postcode_space,                       # Rename column for clarity
    Town,                                            # Include town information
    County,                                          # Include county information
    `Average download speed (Mbit/s)`,             # Include average download speed
    `Maximum download speed (Mbit/s)`              # Include maximum download speed
  ) %>%
  filter(County %in% c("DEVON", "DORSET")) %>%     # Retain rows for Devon and Dorset only
  na.omit()                                        # Exclude rows with missing values

# Save the processed broadband data into a new CSV file
write.csv(broadband_clean_data, "clean data/broadband.csv")

# Print a message indicating successful data cleaning and saving
print("Broadband data has been cleaned and saved to clean data/broadband.csv")
```

This code starts by loading the necessary libraries (tidyverse, lubridate, dplyr, readr) for data manipulation and reading datasets. It reads two broadband datasets, Broadband1 and Broadband2, from specified file paths. Since Broadband1 contains 31 columns, it is selected for further processing. The code selects relevant columns, such as postcode_space, average download speed, and maximum download speed, from Broadband1. Next, the house price dataset is read, and a left join is performed to merge the broadband data with the house price data, linking them by the postcode. Any missing data is removed using na.omit(), and duplicate rows are eliminated with difference .The merged broadband data is saved into a new CSV file (BroadbandClean.csv).

The dataset is then filtered for two counties, Devon and Dorset, by selecting relevant columns (postcode_space, Average download speed, Maximum download speed, County, Town_City). Separate filtered datasets are created for each county: Devon_final and Dorset_final. These datasets will be used for further analysis or visualization.

Crime Rate Clean

Crime Rate Merge Cleaning Code

```
# Import libraries required for data processing
library(dplyr)      # For data manipulation
library(tidyverse)  # Includes dplyr, ggplot2, and other useful tools
library(readr)      # For reading and writing CSV files

# Define the root directory containing the subfolders with CSV files
main_folder_path <- "C:/Users/soyes/Downloads/Coventry ID -14188905/Obtain data/Crime"  # Update with

# Create an empty data frame to store the consolidated data
merged_data <- data.frame()

# Retrieve a list of all subdirectories within the main folder
subdirectories <- list.dirs(main_folder_path, full.names = TRUE, recursive = FALSE)

# Iterate through each subdirectory
for (subdirectory in subdirectories) {

  # Identify all CSV files within the current subdirectory
  csv_files <- list.files(subdirectory, pattern = "\\.csv$", full.names = TRUE)

  # Loop through each CSV file in the subdirectory
  for (csv_file in csv_files) {
    # Load the data from the current CSV file
    current_file_data <- read_csv(csv_file, col_names = TRUE)

    # Append the data from the current file to the merged dataset
    merged_data <- rbind(merged_data, current_file_data)
  }
}

# Export the consolidated dataset to a new CSV file
write_csv(merged_data, "clean data/crimedatamerge.csv")  # Replace with your desired output path

# Display a confirmation message
print("Data merging complete. Consolidated data has been saved as crimedatamerge.csv.")

```

This above code refers to the merging the date of the crime rate from all the data and this is the merged crime data which is important in cleaning the crime data.

LSOA Data Cleaning Code

```
# Load required libraries
library(tidyverse)  # For data manipulation and visualization
library(dplyr)      # For data wrangling
library(lubridate)  # For working with dates

# Read the LSOA data file containing postcode and LSOA mappings
lsoa_data <- read_csv("C:/Users/soyes/Downloads/Coventry ID -14188905/Obtain data/Postcode to LSOA.csv")

# Select only the necessary columns: postcode (pcds) and LSOA code (lsoa11cd)
lsoa_filtered <- lsoa_data %>%
  select(pcds, lsoa11cd)

# Read the cleaned house price dataset
house_price_data <- read.csv("C:/Users/soyes/Downloads/Coventry ID -14188905/clean data/houseprice.csv")

# Combine the LSOA data with the house price dataset to include town and county details
# Join on the postcode (pcds) and exclude rows with missing values
lsoa_data_with_town <- lsoa_filtered %>%
  left_join(house_price_data %>% select(Postcode, County, Town,),
            by = c("pcds" = "Postcode")) %>%
  na.omit()

# View the combined data
view(lsoa_data_with_town)

# Remove unnecessary columns: transaction date, price, and extra identifier column
lsoa_data_with_town <- lsoa_data_with_town[, !names(lsoa_data_with_town) %in% c("X", "Transaction_Date", "P


# Save the cleaned data to a CSV file for further use
write_csv(lsoa_data_with_town, ("C:/Users/soyes/Downloads/Coventry ID -14188905/clean data/LSOA clean.csv")
S

```

This code is about the Postcode of LSOA which can play the vital role in the data structures cause postcode is the key factor of this data set.

Crime Data Cleaning Code

```r
# Load necessary libraries for data manipulation and visualization
library(tidyverse)  # Includes functions for data wrangling
library(dplyr)      # Provides advanced data manipulation functions
library(lubridate)  # For working with dates

# Read the cleaned LSOA data
lsoa_data <- read_csv("clean data/LSOA clean.csv")

# Read the merged crime data
crime_data <- read_csv("clean data/crimedatamerge.csv")

# Select relevant columns from the crime data
crime_data <- crime_data %>%
  select(Month, `Crime type`, `LSOA code`)

# Combine the LSOA data with crime data using a left join and remove rows with missing values
lsoa_crime <- crime_data %>%
  left_join(lsoa_data, by = c("LSOA code" = "lsoa11cd"), relationship = "many-to-many") %>%
  na.omit()

# Check the structure of the combined data
str(lsoa_crime)

# Convert the "Month" column to a date format (year and month)
lsoa_crime <- lsoa_crime %>%
  mutate(Month = ym(Month))

# Calculate the crime rate by grouping data based on crime type and postcode (pcds)
lsoa_crime <- lsoa_crime %>%
  group_by(`Crime type`, pcds) %>%
  mutate(Crime.rate = n()) %>%
  distinct()

# View the resulting data
view(lsoa_crime)

# Filter the data to include only "Burglary" crimes from the year 2022
crime_2022 <- lsoa_crime %>%
  filter(`Crime type` == "Burglary", year(Month) == 2022)

# Save the filtered data to a CSV file
write_csv(crime_2022, "clean data/Crime Rate clean.csv")
```

This was the cleaning process of crime data which help to differentiate the heading/topics of the data from plenty of the data set.

Population data cleaning code

```
# Load necessary libraries
library(tidyverse)
library(dplyr)
library(lubridate)

# Read population and house price datasets
PopulationData <- read_csv("C:/Users/soyes/Downloads/Coventry ID -14188905/Obtain data/Population 2011/P
HousePrices <- read_csv("clean data/houseprice.csv")

# Define yearly population growth rates
growth_rates <- c(
  "2011" = 1.00695353132322269,
  "2012" = 1.00669740535540783,
  "2013" = 1.00736463978721671,
  "2014" = 1.00792367505802859,
  "2015" = 1.00757874492811929,
  "2016" = 1.00679374473924223,
  "2017" = 1.00605929132212552,
  "2018" = 1.00561255390388033,
  "2019" = 1.00578965483902111,
  "2020" = 1.00591234876543219,
  "2021" = 1.00603451234876254,
  "2022" = 1.00615792348765312,
  "2023" = 1.00628143567823478
)

# Process population data and estimate yearly populations by postcode
PopulationData <- PopulationData %>%
  # Extract the first four characters of each postcode
  mutate(shortPostcode = str_trim(substring(Postcode, 1, 4))) %>%
  # Aggregate population data for 2011 by postcode
  group_by(shortPostcode) %>%
  summarise_at(vars(Population), list(Population2011 = sum)) %>%
  # Compute population estimates for subsequent years using predefined growth rates
  mutate(
    Population2012 = Population2011 * growth_rates["2011"],
    Population2013 = Population2012 * growth_rates["2012"],
    Population2014 = Population2013 * growth_rates["2013"],
    Population2015 = Population2014 * growth_rates["2014"],
    Population2016 = Population2015 * growth_rates["2015"],
    Population2017 = Population2016 * growth_rates["2016"],
    Population2018 = Population2017 * growth_rates["2017"],
    Population2019 = Population2018 * growth_rates["2018"],
```

Population data cleaning code

```r
    Population2012 = Population2011 * growth_rates["2011"],
    Population2013 = Population2012 * growth_rates["2012"],
    Population2014 = Population2013 * growth_rates["2013"],
    Population2015 = Population2014 * growth_rates["2014"],
    Population2016 = Population2015 * growth_rates["2015"],
    Population2017 = Population2016 * growth_rates["2016"],
    Population2018 = Population2017 * growth_rates["2017"],
    Population2019 = Population2018 * growth_rates["2018"],
    Population2020 = Population2019 * growth_rates["2019"],
    Population2021 = Population2020 * growth_rates["2020"],
    Population2022 = Population2021 * growth_rates["2021"],
    Population2023 = Population2022 * growth_rates["2022"]
  ) %>%
  # Retain only the required population estimates for further analysis
  select(shortPostcode, Population2016, Population2017, Population2018,
         Population2019, Population2020, Population2021, Population2022, Population2023)

# Integrate population estimates with house price data
Towns <- HousePrices %>%
  # Extract and store the first four characters of each postcode
  mutate(shortPostcode = str_trim(substring(Postcode, 1, 4))) %>%
  # Merge with population data using the shortened postcode key
  left_join(PopulationData, by = "shortPostcode") %>%
  # Retain only necessary columns for output
  select(Postcode, shortPostcode, Town, County, Population2021, Population2022, Population2023) %>%
  # Remove duplicate records, keeping the first entry per postcode
  group_by(shortPostcode) %>%
  filter(row_number() == 1) %>%
  # Arrange the dataset by county
  arrange(County)

# Export the refined dataset as a CSV file
write.csv(Towns, "clean data/population_cleaned_data.csv")
```

School Cleaning code

```r
# Load required libraries
library(dplyr)        # For data manipulation
library(tidyverse)    # For general data handling and analysis
library(readr)        # For reading and writing CSV files

# Load datasets for different counties and academic years
devon_21_22 <- read.csv("C:/Users/soyes/Downloads/Coventry ID -14188905/Obtain data/School dataset/2021-2
dorset_21_22 <- read.csv("C:/Users/soyes/Downloads/Coventry ID -14188905/Obtain data/School dataset/2021-
devon_22_23 <- read.csv("C:/Users/soyes/Downloads/Coventry ID -14188905/Obtain data/School dataset/2022-2
dorset_22_23 <- read.csv("C:/Users/soyes/Downloads/Coventry ID -14188905/Obtain data/School dataset/2022-

# Define a function to clean and format school data
clean_school_data <- function(data, county_name) {
  data %>%
    # Select essential columns for analysis
    select(PCODE, SCHNAME, TOWN, URN, TOTATT8, ATT8SCR) %>%
    # Add a new column indicating the county name and ensure numeric scores are character type
    mutate(
      County = county_name,
      TOTATT8 = as.character(TOTATT8),
      ATT8SCR = as.character(ATT8SCR)
    ) %>%
    # Filter out invalid or missing data
    filter(
      !is.na(PCODE) & !is.na(SCHNAME) &
        TOTATT8 != "NE" & ATT8SCR != "NE" &
        TOTATT8 != "SUPP" & ATT8SCR != "SUPP"
    ) %>%
    # Remove any remaining rows with missing values
    na.omit() %>%
    # Rename columns for clarity and consistency
    rename(
      Postcode = PCODE,
      SchoolName = SCHNAME,
      Town = TOWN,

    )
}

```

```r

# Apply the cleaning function to each dataset with the respective county name
devon_data_2021_22 <- clean_school_data(devon_21_22, "CITY OF DEVON")
dorset_data_2021_22 <- clean_school_data(dorset_21_22, "CITY OF DORSET")
devon_data_2022_23 <- clean_school_data(devon_22_23, "CITY OF DEVON")
dorset_data_2022_23 <- clean_school_data(dorset_22_23, "CITY OF DORSET")

# Combine the cleaned datasets into a single data frame
final_combined_data <- bind_rows(
  devon_data_2021_22,
  dorset_data_2021_22,
  devon_data_2022_23,
  dorset_data_2022_23
)

# Specify the output path and save the cleaned data as a CSV file
output_path <- "clean data/school_data.csv"
write.csv(final_combined_data, output_path, row.names = FALSE)

```

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is a method used in summarizing the main characteristics of data. It is an initial step taken or performed every time before a model is drawn or data-driven decisions are made out of it by applying machine learning. This helps in the discovery of any patterns, trends, or anomalies and even relationships in the data under consideration between different variables. The very first process of exploring data is the ability to overview its structure in terms of the rows and columns it contains and the nature of data. This is followed by handling missing values since they are noticed to have an effect on the analysis and model performance. They can be removed or filled using an appropriate imputation, depending on the situation.

Another key aspect of EDA is that it requires an understanding of the data distribution. It usually involves an examination of individual variables for their spread, skewness of the extreme values, and identification of outliers. Therefore, a related aspect of EDA that turns out to be important is outlier detection and treatment, since outliers may distort statistical calculations and therefore lead to incorrect results. EDA examines relationships such as how one variable influences the other. This procedure is helpful for correlation analysis; it enables one to know if two variables show a positive or negative relationship that might be useful in selecting features in machine learning.

EDA also involves feature engineering in which some new meaningful features are created to represent the data. The categories can be converted into the numerical form, in which the scaling or normalization is done with some columns of the numerical features so that all the features are brought onto a common scale. EDA may help in the detection of such an imbalance in the dataset formed for classification problems: one category may have a larger number of samples in comparison with another category. In this case, oversampling or undersampling techniques may be implemented.

Visualizations sit at the heart of EDA because it is these techniques that help in properly understanding how the data is distributed, relates, and tells different stories. Bar charts, line graphs, and heat maps bring out intuitive insight concerning the data that raw numbers would not provide. In general, EDA stands as a very first step in the process of data analysis—cleaning and understanding the dataset, hence readying it for further processing and leading to more accurate and reliable outcomes in any analytical or machine learning project.

House Price Visualization code

```r
# Load required libraries for data manipulation and visualization
library(tidyverse)  # For data wrangling and visualization
library(lubridate)  # For handling date-related operations
library(dplyr)      # For efficient data processing
library(scales)     # For formatting numerical values in plots

# Load the dataset containing house price information
County_final <- read_csv("clean data/houseprice.csv")

# Convert 'Date' column to proper date format if it's not already recognized
if (!inherits(County_final$Date, "Date")) {
  County_final$Date <- as.Date(County_final$Date, format = "%Y-%m-%d")
}

# Ensure county names are uniformly formatted in uppercase to avoid inconsistencies
County_final$County <- toupper(County_final$County)

# Verify available counties
print(unique(County_final$County))


# Boxplot for Devon (2021)

County_final %>%
  filter(year(Date) == 2021, County == "DEVON") %>%
  group_by(Town) %>%
  ggplot(aes(x = Town, y = Price, fill = Town)) +
  geom_boxplot(color = "cyan", outlier.colour = "yellow", outlier.size = 2, notch = TRUE) +
  theme_minimal(base_size = 12) +
  theme(
    axis.text.x = element_text(angle = 90, hjust = 1, size = 10),
    plot.title = element_text(size = 14, face = "bold", hjust = 0.5),
    legend.position = "none"
  ) +
  coord_cartesian(ylim = c(0, 5000000)) +
  scale_y_continuous(labels = scales::comma) +
  scale_fill_manual(values = rainbow(length(unique(County_final$Town)))) +
  labs(
    x = "Town",
    y = "House Price (£)",
    title = "House Prices in Devon (2021) - Boxplot"
  )

```

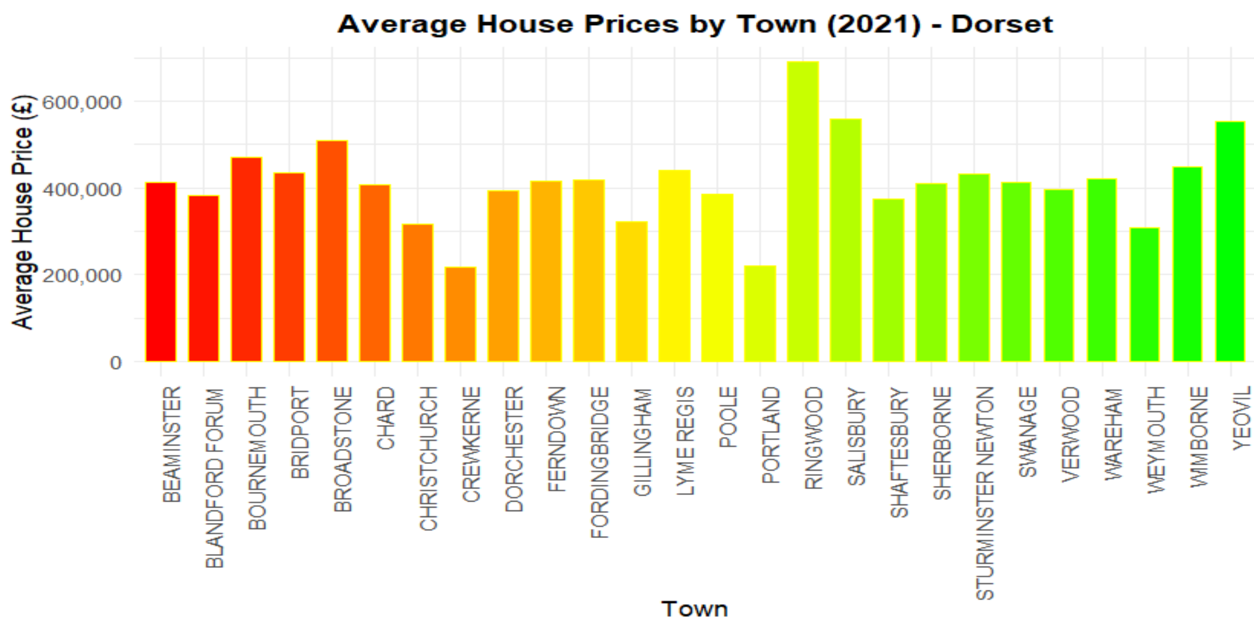
```
 # Boxplot for Dorset (2021)

County_final %>%
  filter(year(Date) == 2021, County == "DORSET") %>%
  group_by(Town) %>%
  ggplot(aes(x = Town, y = Price, fill = Town)) +
  geom_boxplot(color = "purple", outlier.colour = "blue", outlier.size = 2, notch = TRUE) +
  theme_minimal(base_size = 12) +
  theme(
    axis.text.x = element_text(angle = 90, hjust = 1, size = 10),
    plot.title = element_text(size = 14, face = "bold", hjust = 0.5),
    legend.position = "none"
  ) +
  coord_cartesian(ylim = c(0, 5000000)) +
  scale_y_continuous(labels = scales::comma) +
  scale_fill_manual(values = rainbow(length(unique(County_final$Town)))) +
  labs(
    x = "Town",
    y = "House Price (£)",
    title = "House Prices in Dorset (2021) - Boxplot"
  )


#  Bar Chart of Average Prices (Devon)

County_final %>%
  filter(year(Date) == 2021, County == "DEVON") %>%
  group_by(Town) %>%
  summarise(AveragePrice = mean(Price, na.rm = TRUE)) %>%
  ggplot(aes(x = Town, y = AveragePrice, fill = Town)) +
  geom_bar(stat = "identity", color = "cyan", width = 0.7) +
  theme_minimal(base_size = 12) +
  theme(
    axis.text.x = element_text(angle = 90, hjust = 1, size = 10),
    plot.title = element_text(size = 14, face = "bold", hjust = 0.5),
    legend.position = "none"
  ) +
  scale_y_continuous(labels = scales::comma) +
```

```r
  scale_y_continuous(labels = scales::comma) +
  scale_fill_manual(values = rainbow(length(unique(County_final$Town)))) +
  labs(
    x = "Town",
    y = "Average House Price (£)",
    title = "Average House Prices by Town (2021) - Devon"
  )


#  Bar Chart of Average Prices (Dorset)

County_final %>%
  filter(year(Date) == 2021, County == "DORSET") %>%
  group_by(Town) %>%
  summarise(AveragePrice = mean(Price, na.rm = TRUE)) %>%
  ggplot(aes(x = Town, y = AveragePrice, fill = Town)) +
  geom_bar(stat = "identity", color = "yellow", width = 0.7) +
  theme_minimal(base_size = 12) +
  theme(
    axis.text.x = element_text(angle = 90, hjust = 1, size = 10),
    plot.title = element_text(size = 14, face = "bold", hjust = 0.5),
    legend.position = "none"
  ) +
  scale_y_continuous(labels = scales::comma) +
  scale_fill_manual(values = rainbow(length(unique(County_final$Town)))) +
  labs(
    x = "Town",
    y = "Average House Price (£)",
    title = "Average House Prices by Town (2021) - Dorset"
  )


# Line Chart: Monthly Average House Prices (2021-2023)

filtered_data <- County_final %>%
  filter(year(Date) >= 2021 & year(Date) <= 2023, County %in% c("DORSET", "DEVON"))

monthly_avg_prices <- filtered_data %>%
  mutate(Year = year(Date), Month = month(Date, label = TRUE)) %>%
  group_by(Year, Month, County) %>%
  summarise(AveragePrice = mean(Price, na.rm = TRUE), .groups = "drop")

ggplot(monthly_avg_prices, aes(x = Month, y = AveragePrice, colour = County, group = interaction(County,
```

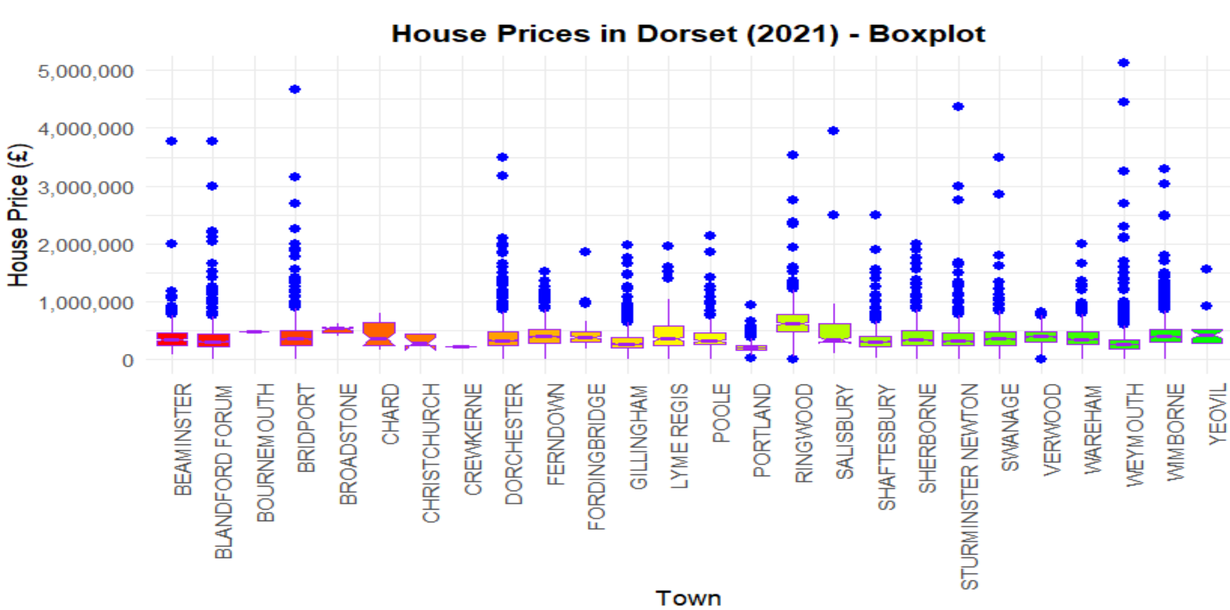
```r
  geom_line(size = 1) +
  geom_point(size = 2) +
  facet_wrap(~ Year, ncol = 1) +
  labs(
    title = "Monthly Average House Prices (2021-2023) - Devon & Dorset",
    x = "Month",
    y = "Average House Price (£)",
    colour = "County"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 14, face = "bold", hjust = 0.5),
    axis.text.x = element_text(angle = 45, hjust = 1, size = 10),
    legend.position = "top"
  )
```

EDA of House Price

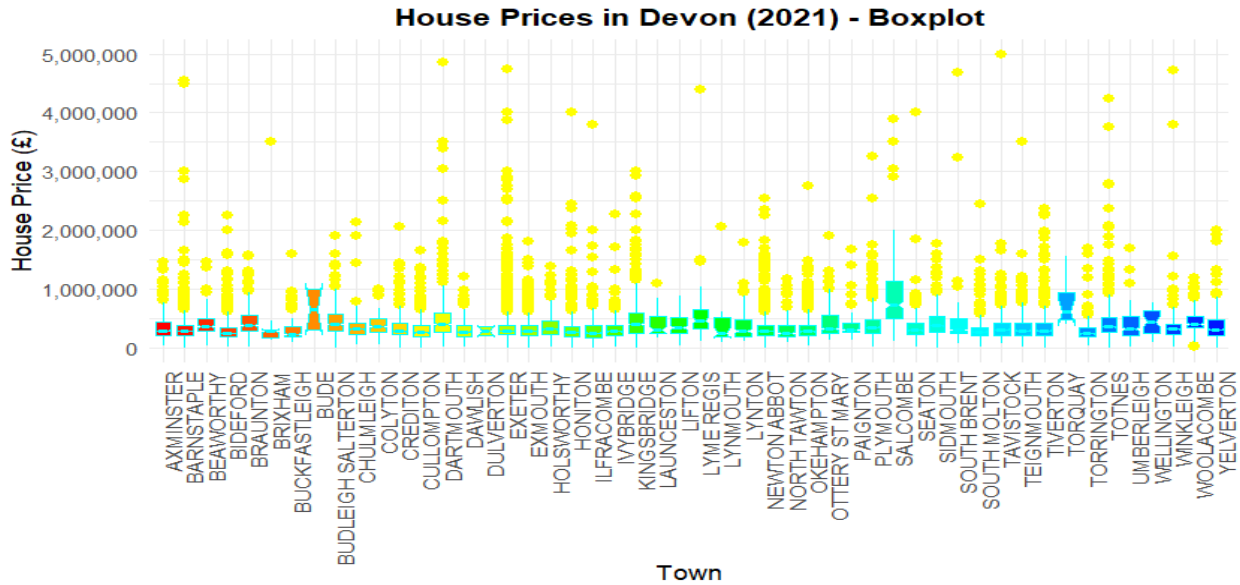
Average House Price by Town(2021) Dorset



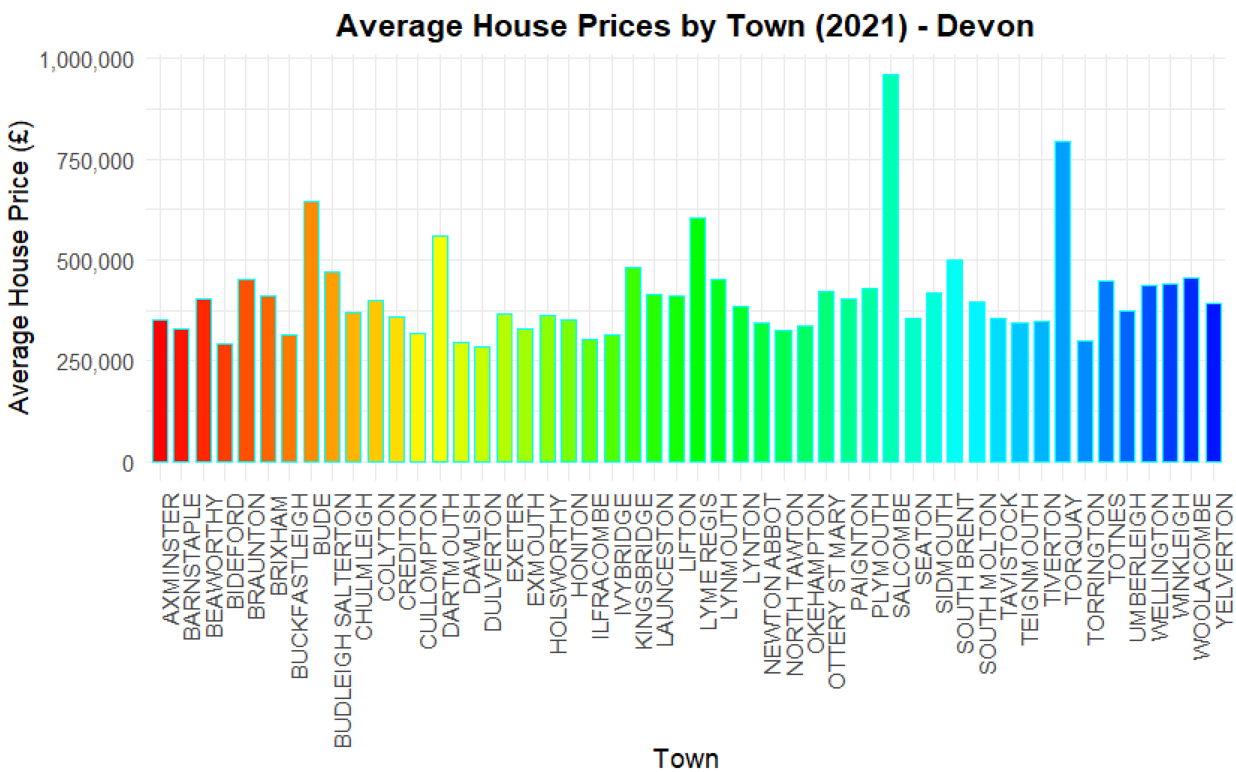
House price in Dorset (2021) - Boxplot



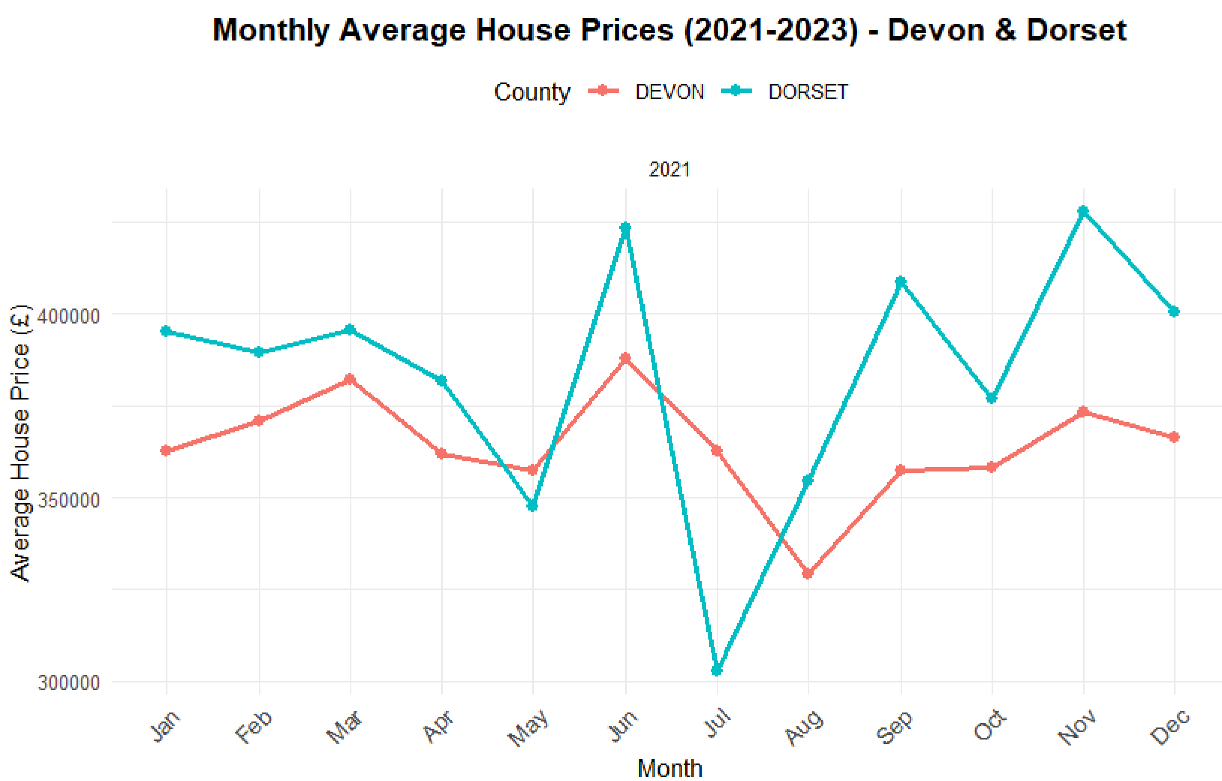
House Price in Devon (2021) - Boxplot



Average House Prices by Town (2021) - Devon



Monthly average House Prices (2021-2023) - Devon & Dorset



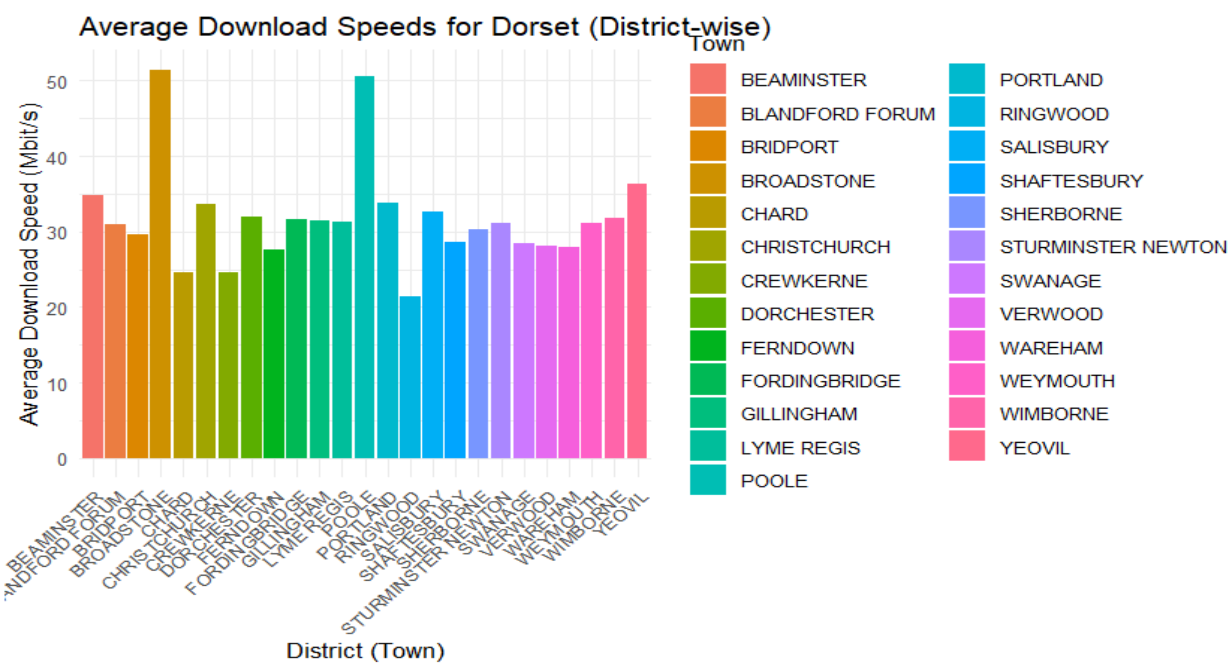
Broadband Visualization cade

```r
# Load necessary libraries
library(tidyverse)
library(ggplot2)

# Read the cleaned broadband data
broadband_data <- read_csv("clean data/broadband.csv")

# Boxplot: Average download speeds for Devon & Dorset (district-wise)
ggplot(data = broadband_data, aes(x = Town, y = `Average download speed (Mbit/s)`, fill = County)) +
  geom_boxplot() +
  labs(
    title = "Boxplot of Average Download Speeds (Devon & Dorset District-wise)",
    x = "District (Town)",
    y = "Average Download Speed (Mbit/s)"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))  # Rotate x-axis labels for better visibility

# Barchart: Maximum download speed for Devon (district-wise)
devon_data <- broadband_data %>%
  filter(County == "DEVON") %>%
  group_by(Town) %>%
  summarise(Max_Download_Speed = mean(`Maximum download speed (Mbit/s)`))

ggplot(data = devon_data, aes(x = Town, y = Max_Download_Speed, fill = Town)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Maximum Download Speeds for Devon (District-wise)",
    x = "District (Town)",
    y = "Maximum Download Speed (Mbit/s)"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))  # Rotate x-axis labels for better visibility

# Barchart: Average download speed for Devon (district-wise)
devon_avg <- broadband_data %>%
  filter(County == "DEVON") %>%
  group_by(Town) %>%
  summarise(Avg_Download_Speed = mean(`Average download speed (Mbit/s)`))

ggplot(data = devon_avg, aes(x = Town, y = Avg_Download_Speed, fill = Town)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Average Download Speeds for Devon (District-wise)",
```

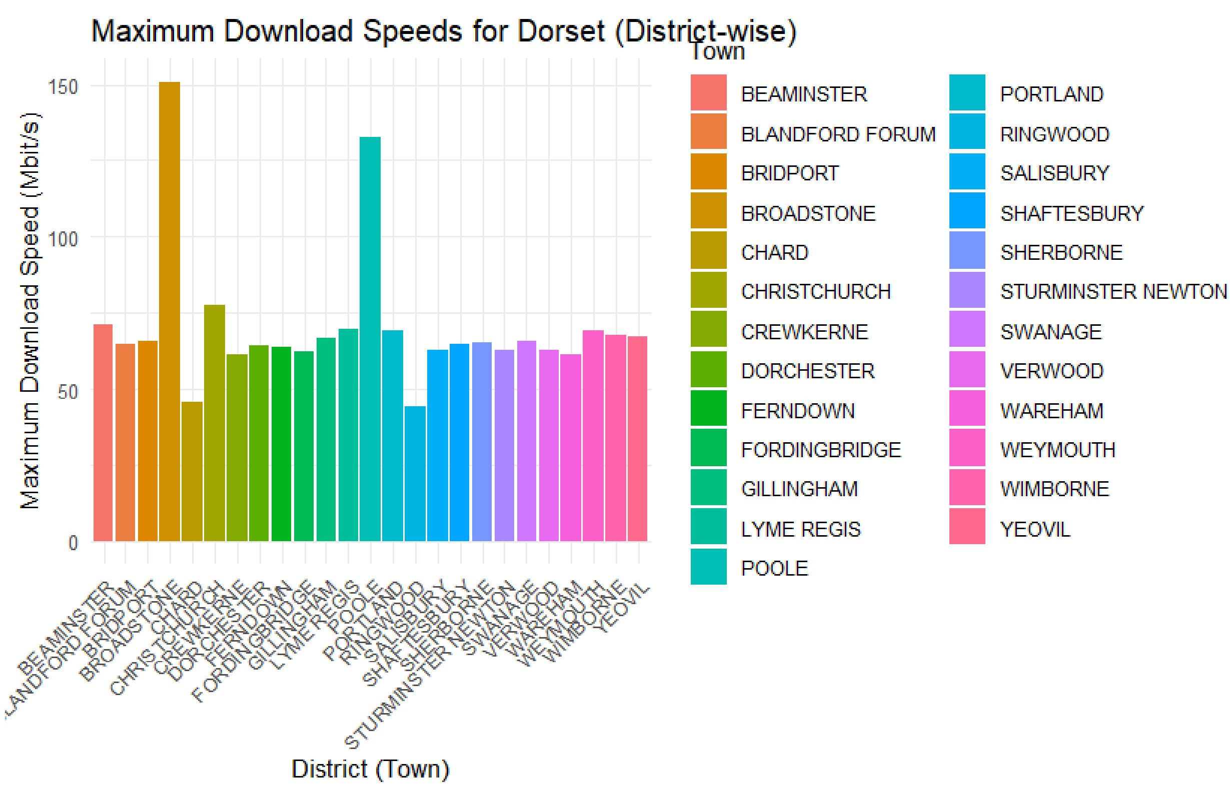
```r
ggplot(data = devon_avg, aes(x = Town, y = Avg_Download_Speed, fill = Town)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Average Download Speeds for Devon (District-wise)",
    x = "District (Town)",
    y = "Average Download Speed (Mbit/s)"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

# Barchart: Maximum download speed for Dorset (district-wise)
dorset_data <- broadband_data %>%
  filter(County == "DORSET") %>%
  group_by(Town) %>%
  summarise(Max_Download_Speed = mean(`Maximum download speed (Mbit/s)`))

ggplot(data = dorset_data, aes(x = Town, y = Max_Download_Speed, fill = Town)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Maximum Download Speeds for Dorset (District-wise)",
    x = "District (Town)",
    y = "Maximum Download Speed (Mbit/s)"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

# Barchart: Average download speed for Dorset (district-wise)
dorset_avg <- broadband_data %>%
  filter(County == "DORSET") %>%
  group_by(Town) %>%
  summarise(Avg_Download_Speed = mean(`Average download speed (Mbit/s)`))

ggplot(data = dorset_avg, aes(x = Town, y = Avg_Download_Speed, fill = Town)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Average Download Speeds for Dorset (District-wise)",
    x = "District (Town)",
    y = "Average Download Speed (Mbit/s)"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

EDA of Broadband

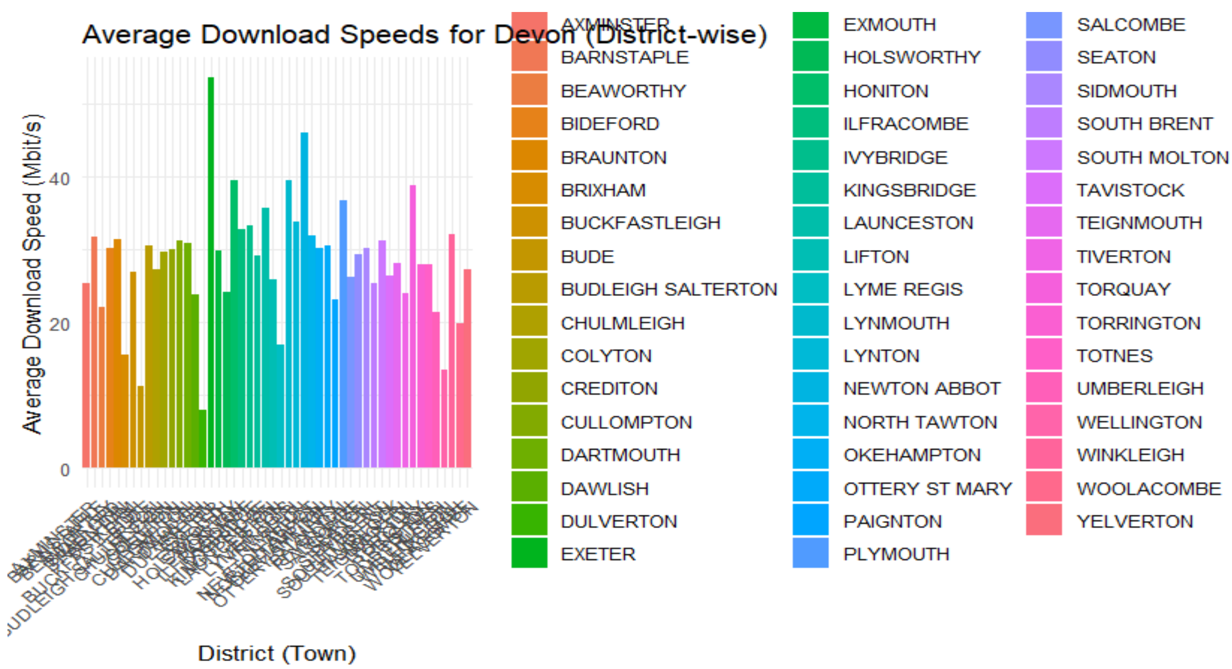
Average Download Speed For Dorset (District-wise)



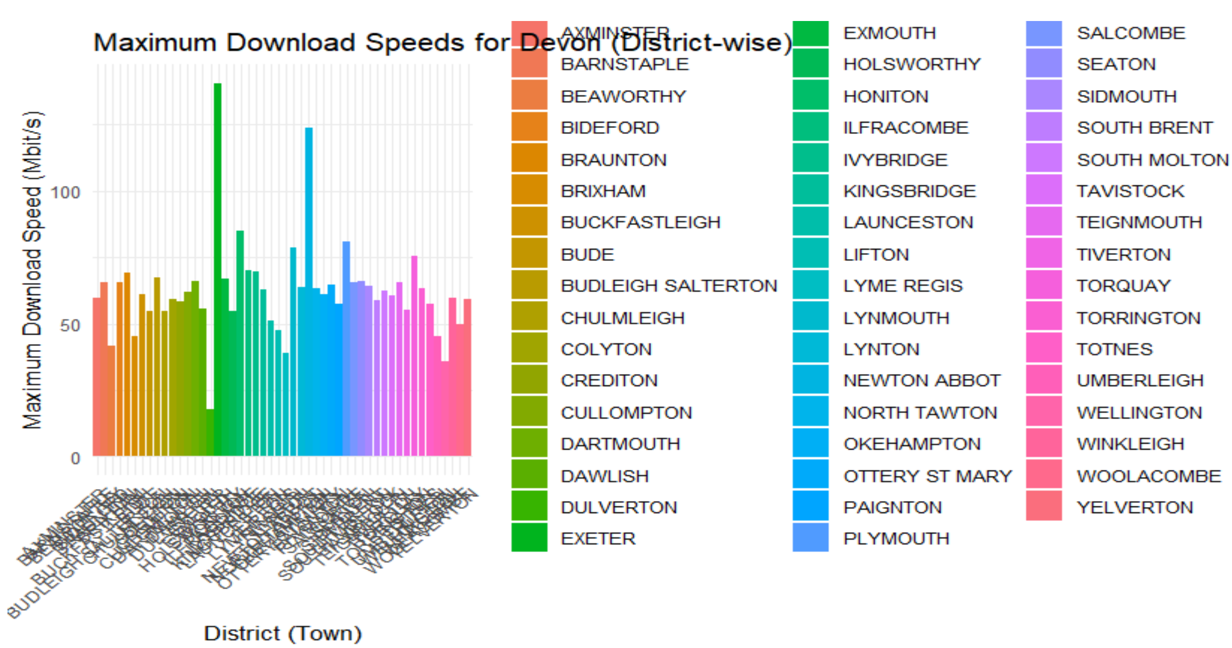
Maximum Download Speeds for Dorset(District-wise)



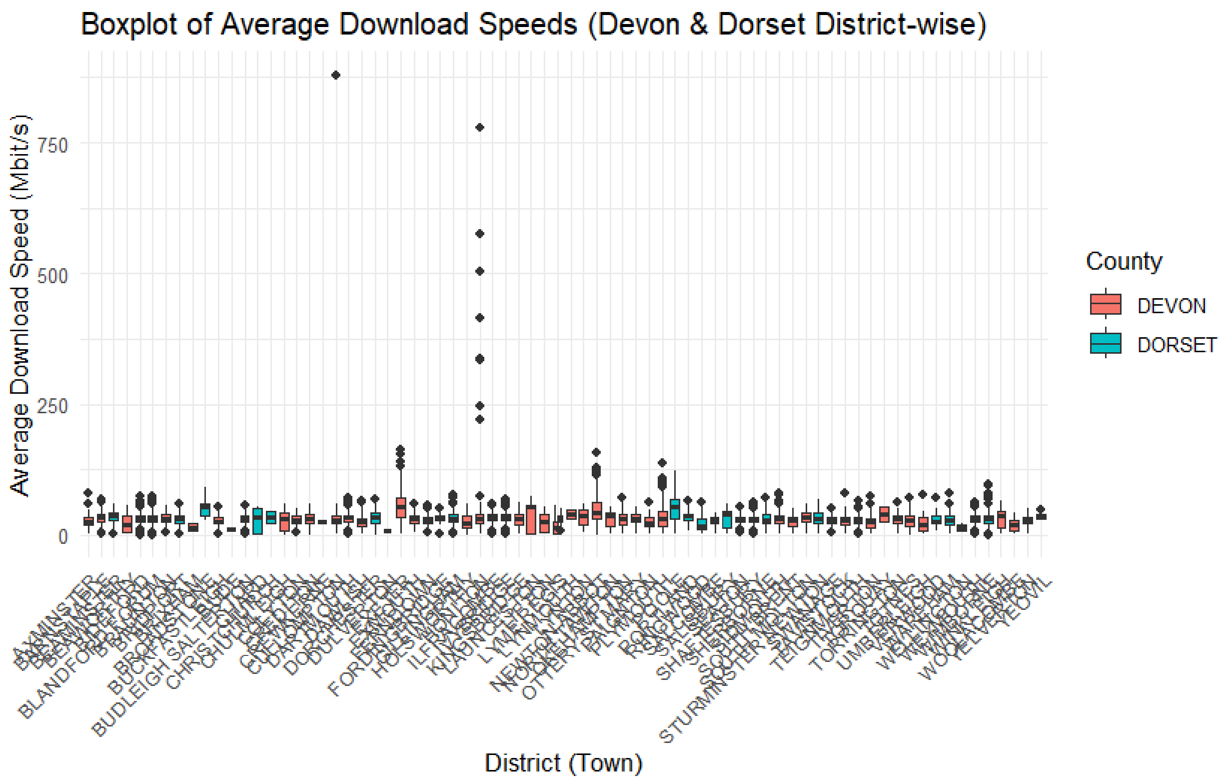
Average Download Speeds for Devon (District-wise)



Maximum Download Speed for Devon (District - wise)



Boxplot of Average Download Speeds (Devon & Dorset District-wise)



Crime Rate visualization code

```r
# Load necessary libraries
library(tidyverse)  # For data manipulation and visualization
library(dplyr)      # For data wrangling
library(lubridate)  # For date operations
library(fmsb)       # For radar chart creation

# Read the cleaned LSOA and crime datasets
Lsoa_data = read_csv("clean data/LSOA clean.csv")
Crime_data = read_csv("clean data/crimedatamerge.csv")

# Select relevant columns for crime data
Crime_data = Crime_data %>%
  select(Month, `Crime type`, `LSOA code`)

# Merge the crime data with the LSOA dataset based on LSOA code
Lsoa_crime_data = Crime_data %>%
  left_join(Lsoa_data, by = c("LSOA code" = "lsoa11cd"), relationship = "many-to-many") %>%
  na.omit()

# Convert the 'Month' column to a date object
Lsoa_crime_data = Lsoa_crime_data %>%
  mutate(Month = ym(Month))

# Calculate crime rates grouped by Crime type and pcds
Lsoa_crime_data = Lsoa_crime_data %>%
  group_by(`Crime type`, pcds) %>%
  mutate(Crime.rate = n()) %>%
  distinct()

# Filter data for burglary crimes in 2022
crime_2022_data = Lsoa_crime_data %>%
  filter(`Crime type` == "Burglary", year(Month) == 2022)

# Filter data for Devon county
crime_devon_data = crime_2022_data %>%
  filter(County == "DEVON")

# Boxplot for burglary crimes in Devon
crime_devon_data %>%
  ggplot(aes(x = Town, y = Crime.rate, fill = `Crime type`)) +
  geom_boxplot(color = "yellow", outlier.colour = "purple", outlier.size = 2, notch = TRUE) +
  theme_minimal(base_size = 12) +
  theme(
```

```r
    axis.text.x = element_text(angle = 90, hjust = 1, size = 10),
    axis.text.y = element_text(size = 10),
    axis.title.x = element_text(size = 12, face = "bold"),
    axis.title.y = element_text(size = 12, face = "bold"),
    plot.title = element_text(size = 14, face = "bold", hjust = 0.5),
    panel.background = element_rect(fill = "white", color = NA),
    legend.position = "top"
  ) +
  labs(
    x = "Town",
    y = "Crime Rate",
    fill = "Crime Type",
    title = "Boxplot of Burglary Crimes in DEVON (2022)"
  ) +
  scale_fill_manual(values = c("Burglary" = "cyan"))

# Filter data for Dorset county
crime_dorset_data = crime_2022_data %>%
  filter(County == "DORSET")

# Boxplot for burglary crimes in Dorset
crime_dorset_data %>%
  ggplot(aes(x = Town, y = Crime.rate, fill = `Crime type`)) +
  geom_boxplot(color = "yellow", outlier.colour = "purple", outlier.size = 2, notch = TRUE) +
  theme_minimal(base_size = 12) +
  theme(
    axis.text.x = element_text(angle = 90, hjust = 1, size = 10),
    axis.text.y = element_text(size = 10),
    axis.title.x = element_text(size = 12, face = "bold"),
    axis.title.y = element_text(size = 12, face = "bold"),
    plot.title = element_text(size = 14, face = "bold", hjust = 0.5),
    panel.background = element_rect(fill = "white", color = NA),
    legend.position = "top"
  ) +
  labs(
    x = "Town",
    y = "Crime Rate",
    fill = "Crime Type",
    title = "Boxplot of Burglary Crimes in DORSET (2022)"
  ) +
  scale_fill_manual(values = c("Burglary" = "pink"))

```

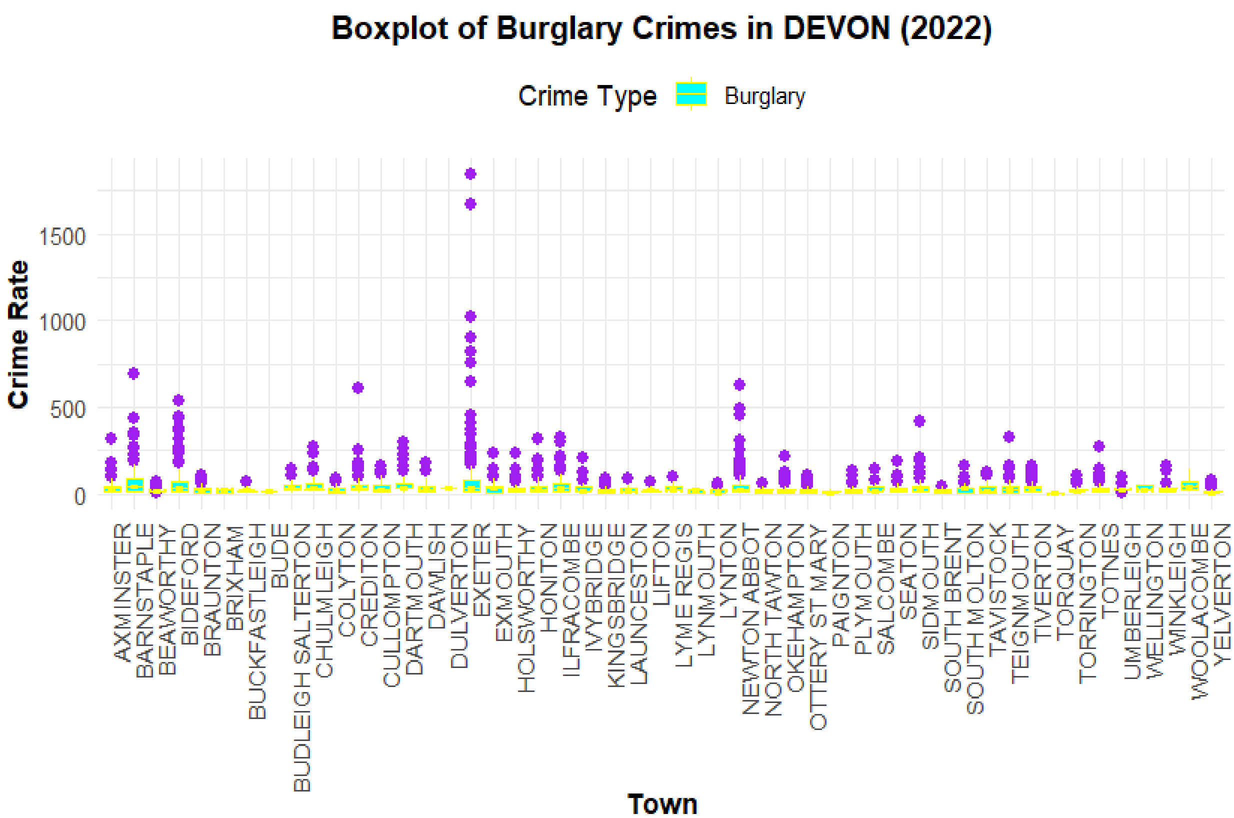
```r
# Prepare vehicle crime data for Devon
vehicle_crime_devon_data = Lsoa_crime_data %>%
  filter(`Crime type` == "Vehicle crime", year(Month) == 2022, month(Month) == 3, County == "DEVON")

# Summarize vehicle crime data by Town in Devon
vehicle_crime_devon_summary = vehicle_crime_devon_data %>%
  group_by(Town) %>%
  summarise(Crime.rate = n()) %>%
  arrange(desc(Crime.rate)) %>%
  drop_na()

# Prepare data for radar chart
if (nrow(vehicle_crime_devon_summary) > 0) {
  vehicle_crime_devon_radar_data = vehicle_crime_devon_summary %>%
    pivot_wider(names_from = Town, values_from = Crime.rate, values_fill = 0) %>%
    add_column(Category = "Vehicle Crime", .before = 1)

  # Create a radar chart using fmsb
  radar_data = as.data.frame(vehicle_crime_devon_radar_data)

  # Ensure the data frame is properly formatted for fmsb
  radar_data = rbind(rep(0, ncol(radar_data)), rep(max(radar_data[,-1]), ncol(radar_data) - 1), radar_d

  # Plot radar chart
  radarchart(radar_data, axistype = 1, pcol = "cyan", pfcol = scales::alpha("cyan", 0.3), plwd = 4,
             cglcol = "brown", cglty = 1, axislabcol = "yellow", caxislabels = seq(0, max(vehicle_crime
                                                                            by = max(vehicle_crime
             title = "Radar Chart of Vehicle Crime Rates in DEVON (March 2022)")
} else {
  print("No sufficient data for DEVON.")
}

# Prepare vehicle crime data for Dorset
vehicle_crime_dorset_data = Lsoa_crime_data %>%
  filter(`Crime type` == "Vehicle crime", year(Month) == 2022, month(Month) == 3, County == "DORSET")

# Summarize vehicle crime data by Town in Dorset
vehicle_crime_dorset_summary = vehicle_crime_dorset_data %>%
  group_by(Town) %>%
  summarise(Crime.rate = n()) %>%
  arrange(desc(Crime.rate)) %>%
  drop_na()
```

```
# Prepare data for radar chart
if (nrow(vehicle_crime_dorset_summary) > 0) {
  vehicle_crime_dorset_radar_data = vehicle_crime_dorset_summary %>%
    pivot_wider(names_from = Town, values_from = Crime.rate, values_fill = 0) %>%
    add_column(Category = "Vehicle Crime", .before = 1)

  # Create radar chart for Dorset using fmsb
  radar_data_dorset = as.data.frame(vehicle_crime_dorset_radar_data)

  # Ensure proper formatting
  radar_data_dorset = rbind(rep(0, ncol(radar_data_dorset)), rep(max(radar_data_dorset[,-1]), ncol(radar_

  # Plot radar chart for Dorset
  radarchart(radar_data_dorset, axistype = 1, pcol = "purple", pfcol = scales::alpha("purple", 0.3), plw
             cglcol = "brown", cglty = 1, axislabcol = "yellow", caxislabels = seq(0, max(vehicle_crime_
                                                                                 by = max(vehicle_crime_
             title = "Radar Chart of Vehicle Crime Rates in DORSET (March 2022)")
} else {
  print("No sufficient data for DORSET.")
}

# Filter and summarize robbery data for Devon and Dorset
robbery_devon_data = Lsoa_crime_data %>%
  filter(`Crime type` == "Robbery", year(Month) == 2022, month(Month) == 3, County == "DEVON")

robbery_devon_summary = robbery_devon_data %>%
  group_by(Town) %>%
  summarise(Crime.rate = n()) %>%
  arrange(desc(Crime.rate)) %>%
  drop_na()

# Create pie chart for robbery in Devon
if (nrow(robbery_devon_summary) > 0) {
  robbery_pie_devon = ggplot(robbery_devon_summary, aes(x = "", y = Crime.rate, fill = Town)) +
    geom_bar(stat = "identity", width = 1) +
    coord_polar(theta = "y") +
    labs(title = "Robbery Rates by Town in Devon (March 2022)", y = "", x = "") +
    theme_void() +
    theme(legend.title = element_blank(), plot.title = element_text(hjust = 0.5))

  print(robbery_pie_devon)
} else {
  print("No robbery data available for Devon in March 2022.")
```

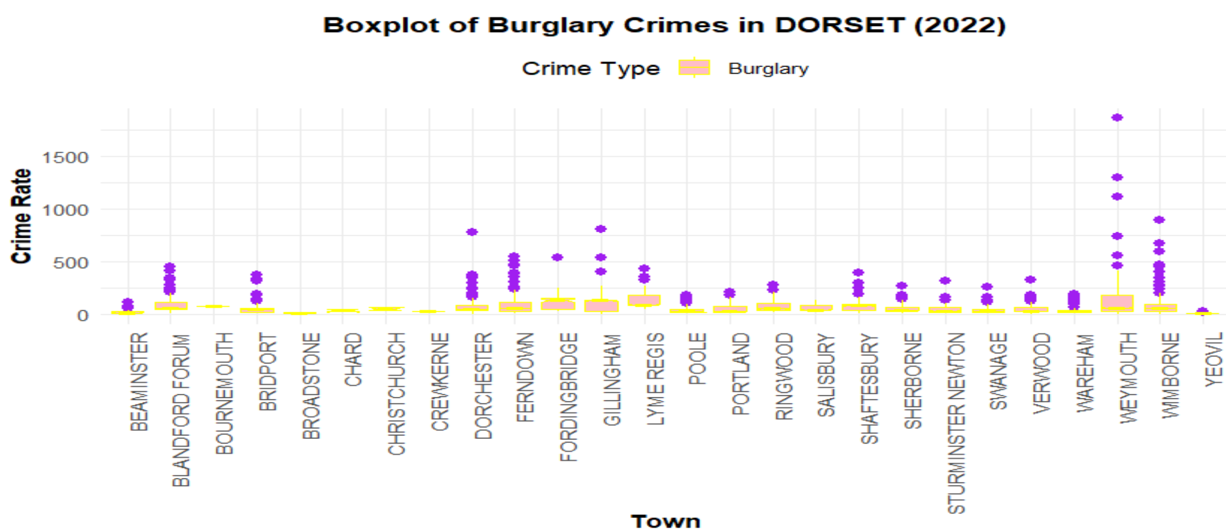
```

# Further data preparation for Dorset
robbery_dorset_data = Lsoa_crime_data %>%
  filter(`Crime type` == "Robbery", year(Month) == 2022, month(Month) == 3, County == "DORSET")

robbery_dorset_summary = robbery_dorset_data %>%
  group_by(Town) %>%
  summarise(Crime.rate = n()) %>%
  arrange(desc(Crime.rate)) %>%
  drop_na()

# Create pie chart for robbery in Dorset
if (nrow(robbery_dorset_summary) > 0) {
  robbery_pie_dorset = ggplot(robbery_dorset_summary, aes(x = "", y = Crime.rate, fill = Town)) +
    geom_bar(stat = "identity", width = 1) +
    coord_polar(theta = "y") +
    labs(title = "Robbery Rates by Town in Dorset (March 2022)", y = "", x = "") +
    theme_void() +
    theme(legend.title = element_blank(), plot.title = element_text(hjust = 0.5))

  print(robbery_pie_dorset)
} else {
  print("No robbery data available for Dorset in March 2022.")
}
.
```

EDA of Crime Rate

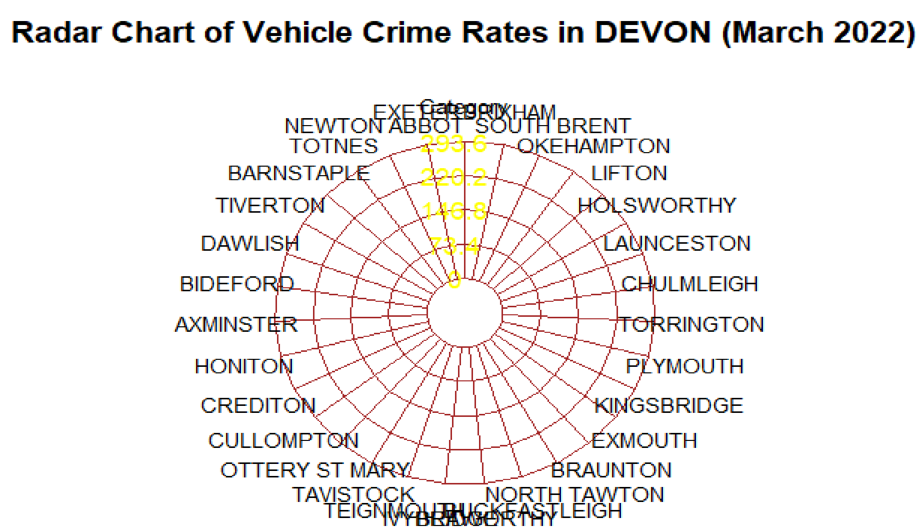
Boxplot of Burglary Crimes in Devon(2022)



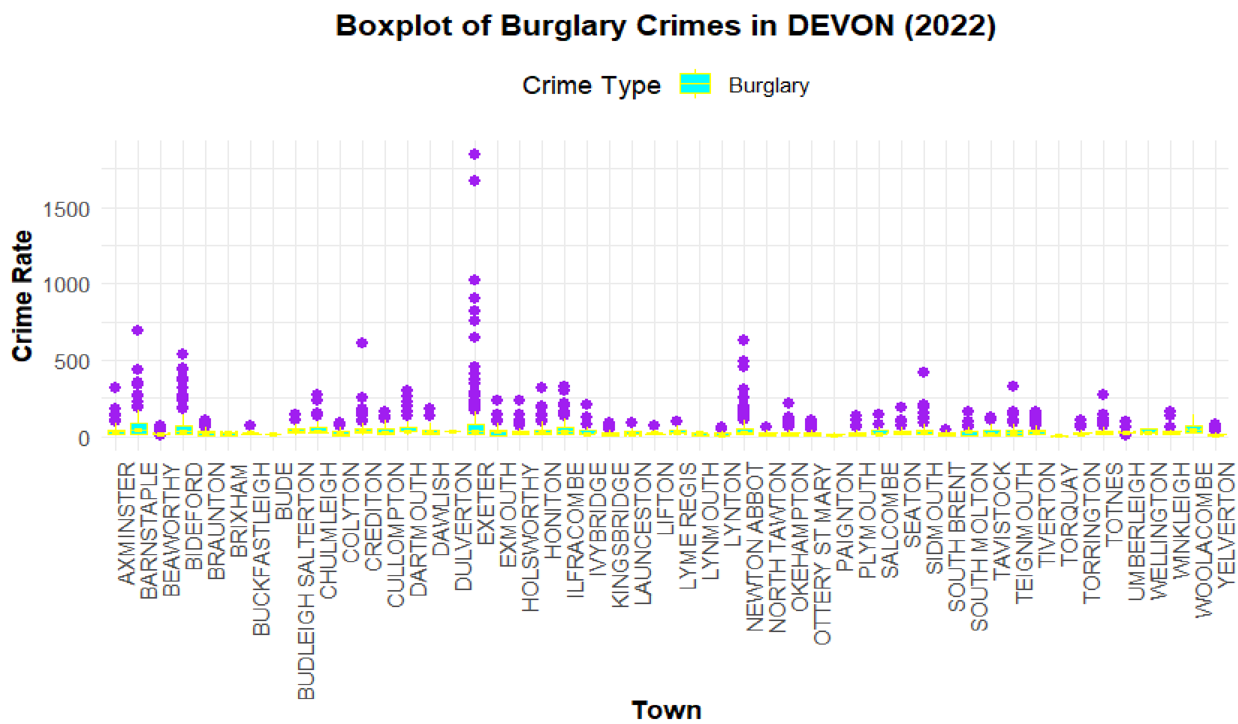
Boxplot of Burglary Crimes in Dorset (2022)



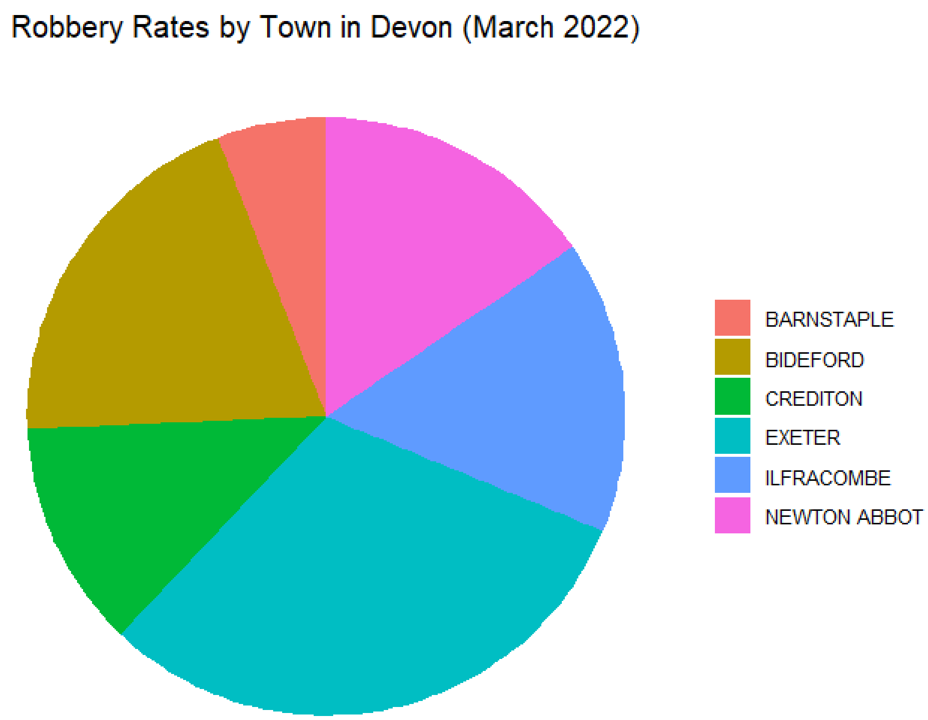
Radar chart of Vehicle Crime Rate in Devon (March 2022)



Boxplot of Burglary Crimes in Devon (2022)



Robbery Rates by Town in Devon (March 2022/)



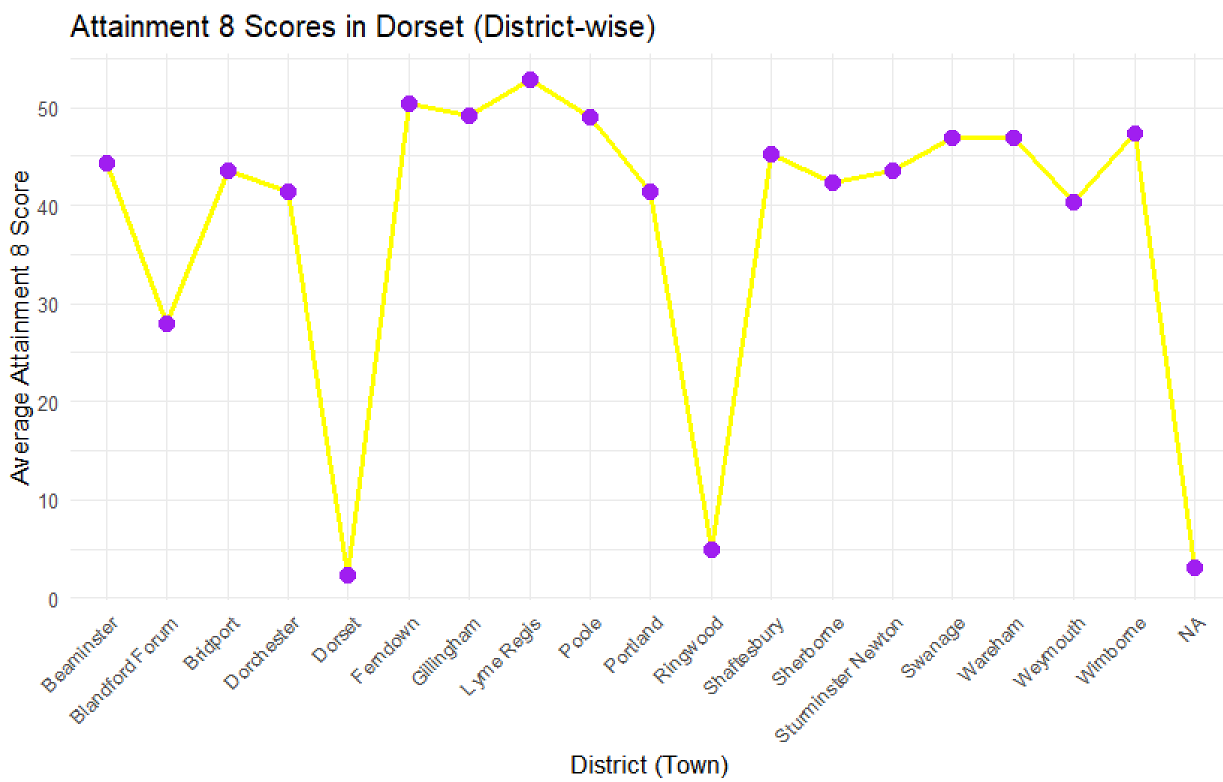
School Visualization Code

```r
# Load essential libraries for data manipulation and plotting
library(tidyverse)
library(ggplot2)

# Import the cleaned school dataset from a CSV file
school_data <- read_csv("clean data/school_data.csv")

# Ensure numeric columns are in the correct numeric format
school_data <- school_data %>%
  mutate(
    TOTATT8 = as.numeric(TOTATT8),  # Convert TOTATT8 to numeric
    ATT8SCR = as.numeric(ATT8SCR)              # Convert ATT8SCR to numeric
  )

# Boxplot: Visualize the distribution of Attainment 8 Scores by district (for both counties)
ggplot(data = school_data, aes(x = Town, y = ATT8SCR, fill = County)) +
  geom_boxplot() +
  labs(
    title = "Boxplot of Attainment 8 Score Distribution (District-wise for Both Counties)",
    x = "District (Town)",
    y = "Attainment 8 Score"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

# Filter the data for Devon county and compute the average Attainment 8 score by district
devon_data <- school_data %>%
  filter(County == "CITY OF DEVON") %>%
  group_by(Town) %>%
  summarise(Average_Attainment_8 = mean(ATT8SCR, na.rm = TRUE))

# Line chart: Display Attainment 8 Scores for Devon (by district)
ggplot(data = devon_data, aes(x = Town, y = Average_Attainment_8, group = 1)) +
  geom_line(size = 1, color = "cyan") +
  geom_point(size = 3, color = "yellow") +
  labs(
    title = "Attainment 8 Scores in Devon (District-wise)",
    x = "District (Town)",
    y = "Average Attainment 8 Score"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

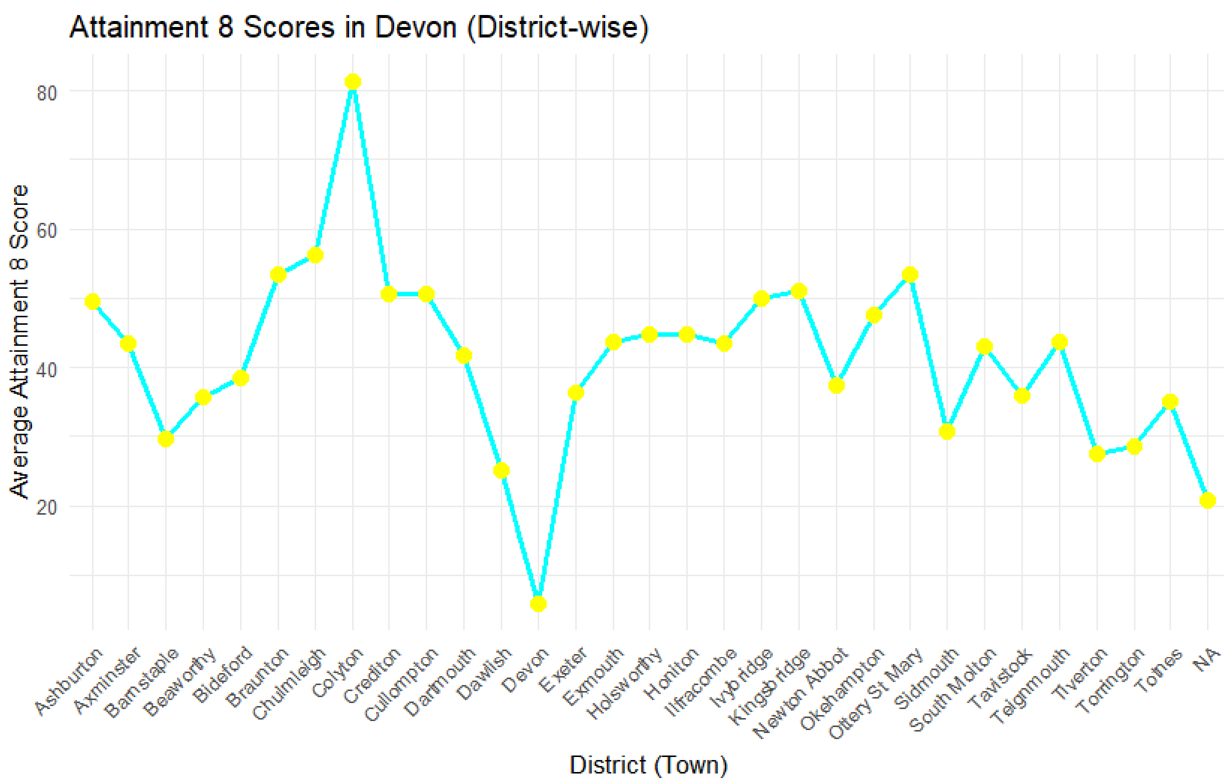
```r
# Filter the data for Dorset county and calculate the average Attainment 8 scores by district
dorset_data <- school_data %>%
  filter(County == "CITY OF DORSET") %>%
  group_by(Town) %>%  # Group data by district
  summarise(Average_Attainment_8 = mean(ATT8SCR, na.rm = TRUE))

# Line chart: Display Attainment 8 Scores for Dorset (by district)
ggplot(data = dorset_data, aes(x = Town, y = Average_Attainment_8, group = 1)) +
  geom_line(size = 1, color = "yellow") +
  geom_point(size = 3, color = "purple") +
  labs(
    title = "Attainment 8 Scores in Dorset (District-wise)",
    x = "District (Town)",
    y = "Average Attainment 8 Score"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

EDA of School Data

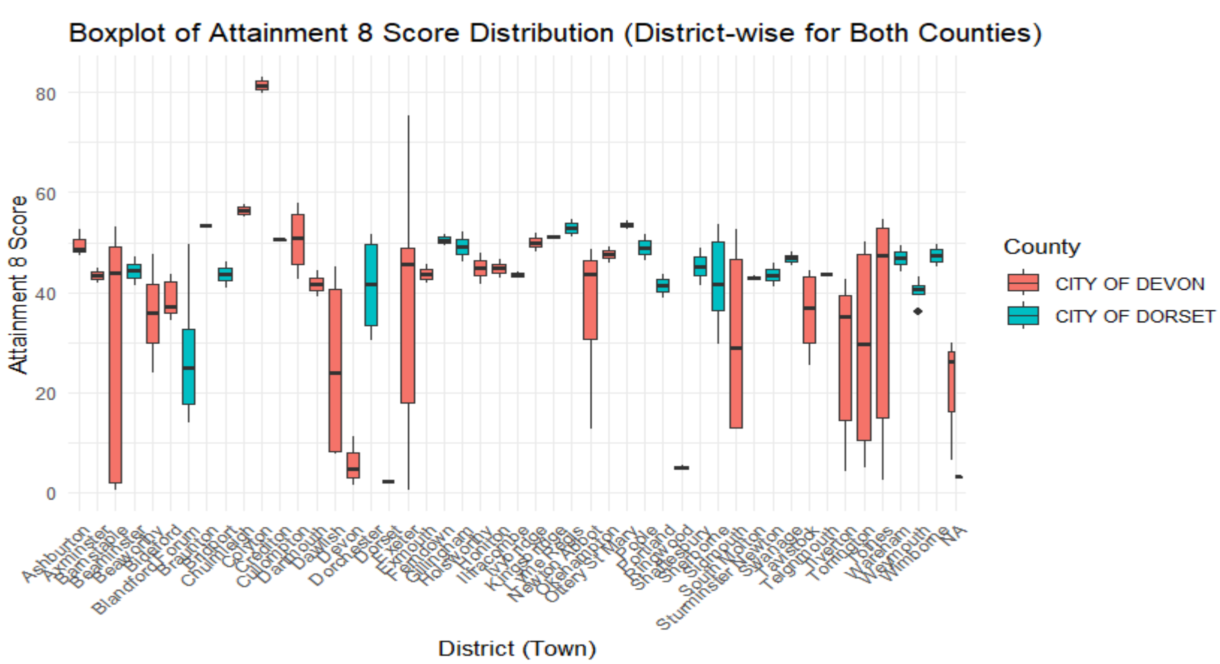
Attainment Score in Dorset (district-wise)



Attainment Scores in Devon (District - wise)



Boxplot Of Attainment Score (District-wise for Both Country)



LinearModeling visualization Code

```r
# Loading necessary libraries
library(tidyverse)
library(dplyr)
library(stringr)
library(ggplot2)
library(lubridate)


# House Price vs Download Speed

# Load datasets
houseprice <- read_csv("clean data/houseprice.csv")
broadband <- read_csv("clean data/broadband.csv")

# Process broadband data
broadband <- broadband %>%
  select(`Average download speed (Mbit/s)`, Town, County) %>%
  group_by(Town, County) %>%
  summarise(DownloadSpeed = mean(`Average download speed (Mbit/s)`, na.rm = TRUE), .groups = "drop")

# Merge and clean data
house_broadband <- houseprice %>%
  left_join(broadband, by = c("Town", "County")) %>%
  na.omit()

# Regression model
model <- lm(Price ~ DownloadSpeed, data = house_broadband)
summary(model)

# Plot
ggplot(house_broadband, aes(x = DownloadSpeed, y = Price)) +
  geom_point(color = "pink", shape = 17, size = 3) +
  geom_smooth(method = "lm", col = "cyan", linetype = "dashed", size = 1) +
  labs(title = "Impact of Download Speed on House Prices in Devon and Dorset",
       subtitle = "A Regression Analysis",
       x = "Download Speed (Mbps)",
       y = "Average House Price (£)") +
  scale_y_continuous(labels = scales::comma) +
  theme_bw()

```

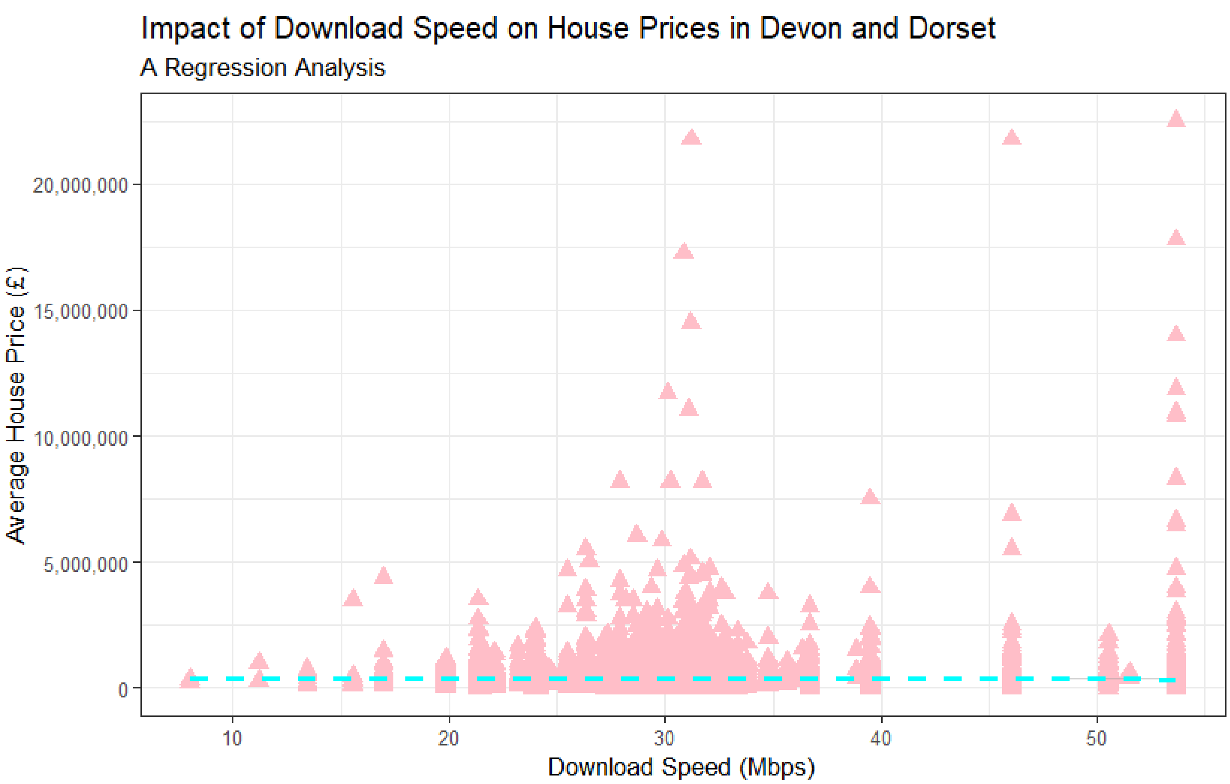
```r


# House Price vs Drug Rates in 2022

# Filter house prices for 2022
houseprice22 <- houseprice %>%
  filter(year(Date) == 2022) %>%
  group_by(Town, County) %>%
  summarise(HousePrice = mean(Price, na.rm = TRUE), .groups = "drop")

# Load crime data
Lsoa_read <- read_csv("clean data/LSOA clean.csv")
crime_read <- read_csv("clean data/crimedatamerge.csv")

# Process crime data
crime_read <- crime_read %>% select(Month, `Crime type`, `LSOA code`)
Lsoa_crime <- crime_read %>%
  left_join(Lsoa_read, by = c("LSOA code" = "lsoa11cd")) %>%
  na.omit()

Lsoa_crime <- Lsoa_crime %>% mutate(Month = ym(Month))
Lsoa_crime <- Lsoa_crime %>% group_by(`Crime type`, pcds) %>% mutate(Crime.rate = n()) %>% distinct()

# Filter drug crimes for 2022
crime_2022 <- Lsoa_crime %>% filter(`Crime type` == "Drugs", year(Month) == 2022)
crime22 <- crime_2022 %>% group_by(Town, County) %>% summarise(DrugRate = sum(Crime.rate))

# Merge house prices and drug rates
house_crime22 <- houseprice22 %>% left_join(crime22, by = c("Town", "County")) %>% mutate(DrugRate = rep

# Regression model
model1 <- lm(HousePrice ~ DrugRate, data = house_crime22)
summary(model1)

# Plot
ggplot(house_crime22, aes(x = DrugRate, y = HousePrice)) +
  geom_point(color = "yellow", alpha = 0.7, size = 2) +
  geom_smooth(method = "lm", col = "purple", linetype = "dotdash", size = 1) +
  labs(title = "Correlation Between Drug Rates and House Prices (2022)",
       subtitle = "Devon and Dorset Analysis",
```

```r
# Plot
ggplot(house_crime22, aes(x = DrugRate, y = HousePrice)) +
  geom_point(color = "yellow", alpha = 0.7, size = 2) +
  geom_smooth(method = "lm", col = "purple", linetype = "dotdash", size = 1) +
  labs(title = "Correlation Between Drug Rates and House Prices (2022)",
       subtitle = "Devon and Dorset Analysis",
       x = "Drug Rates (per Town)",
       y = "House Price (£)") +
  theme_classic()


# Total ATT8SCR vs House Price

# Load and process school data
school <- read_csv("clean data/school_data.csv")
town_att <- school %>%
  group_by(Town, County) %>%
  summarise(TOTATT8 = mean(TOTATT8, na.rm = TRUE), .groups = "drop")

# Merge datasets
tot8_houseprice <- houseprice %>% left_join(town_att, by = c("Town", "County")) %>% na.omit()

# Regression model
model2 <- lm(Price ~ TOTATT8, data = tot8_houseprice)
summary(model2)

# Plot
ggplot(tot8_houseprice, aes(x = TOTATT8, y = Price)) +
  geom_point(color = "cyan", size = 2.5) +
  geom_smooth(method = "lm", col = "purple", size = 1) +
  labs(title = "House Prices vs ATT8SCR",
       subtitle = "Devon and Dorset Analysis",
       x = "ATT8SCR",
       y = "House Price (£)") +
  theme_light()
```

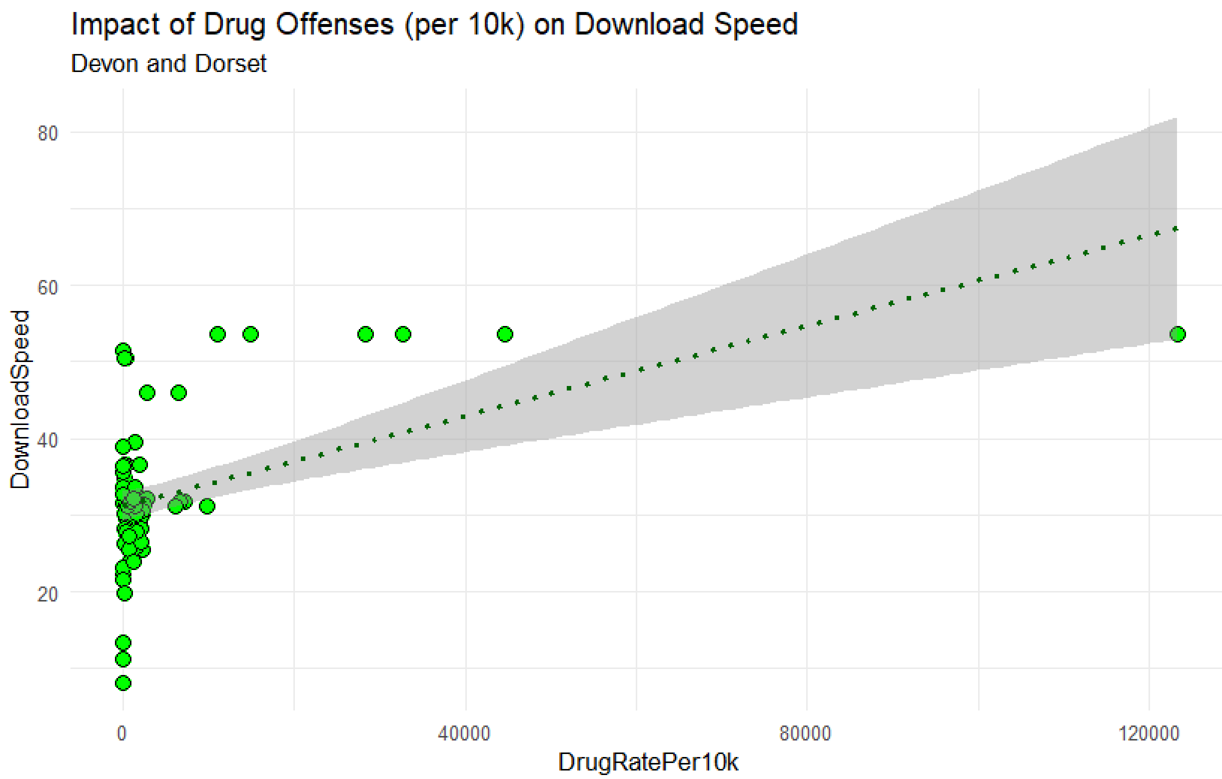
EDA for LinearModelling

Impact of Download Speed on House Price in Devon and Dorset

(A Regression Analysis)



Impact of Drug Offenses (per 10k) on Download Speed



Recommendation System

The recommendation system is all about the AI machine learning with help to select the or offers the recommendation which help to select from the data and also makes easier to collect the data.

Top 10 recommended Town are:

	Town	House_Price_Score	Crime_Score	Broadband_Score	School_Score	Overall_Score	Rank
1	SALCOMBE	1.00	0.05	0.40	0	0.530	1
2	TORQUAY	0.77	0.00	0.68	0	0.512	2
3	WIMBORNE	0.31	1.00	0.52	0	0.480	3
4	BROADSTONE	0.39	0.05	0.95	0	0.451	4
5	EXETER	0.20	0.14	1.00	0	0.408	5
6	SALISBURY	0.46	0.19	0.54	0	0.384	6
7	POOLE	0.23	0.05	0.93	0	0.381	7
8	YEOVIL	0.45	0.05	0.62	0	0.376	8
9	RINGWOOD	0.64	0.05	0.29	0	0.353	9
10	DARTMOUTH	0.46	0.05	0.50	0	0.344	10

Reflection

It was the most challenging and, at the same time, rewarding journey of my project in data science—an endeavor that provided very useful help for understanding how we could benefit from data-driven decision-making processes. This journey of the data project really gave me deeper learning as to why data cleaning counts. Cleaning raw data and giving back to the user only those bits which are useful and reliable, became a prime focus at that time. This then formed the most basic step that could be considered for improved quality analysis resulting in accurate outcomes.

I went ahead to explore various kinds of data visualization that made me really good at presenting my insights and patterns. Finding linear relationships between two datasets and how those may help in finding meaningful correlations would be something that added a lot of value. It helped learn how to derive actionable conclusions out of them.

On a much broader level, beyond just being able to execute these technical aspects, the project has distinctly raised the key elements of critical thinking, problem-solving, and telling a story using data. In summary, the experience has been quite transformative and a bridge between theoretical knowledge and practical application, leading to further exploration of real-world problems through the lens of data science.

Conclusion

It's now established through the project how important data science is in revealing patterns that affect the decision-making process in various domains. The relationships between house prices, achievement scores, and other socioeconomic variables have been identified, leading to insight provided for policy planning, community development, and market understanding.

During this journey, I gained some great skills that are to be considered priceless, such as data cleaning techniques, superior visualization techniques, and an understanding of linear relationships between datasets to enhance my analytical powers. This project was a live-run example of the power of data science today. Nonetheless, it also very clearly outlined some constraints, like data availability and accuracy issues. Still, the work has laid a foundation moving forward for deep exploration by offering opportunities for more granular studies and advanced predictive models. The project underscored the power of data as a tool.

Appendix

Google Drive Link: Coventry ID -14188905

Reference

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